

Davide Scassola

Applied Scientist · PhD in Artificial Intelligence

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Applied scientist specializing in AI with a strong background in computer science, mathematics, and statistics. Experienced in developing and training machine learning models from scratch, with publications on generative AI in top-tier venues (AAAI, UAI). Industry experience spans collaborative software engineering, statistical analysis, and research.

Education

University of Trieste

Trieste, Italy

Doctorate (PhD) in Applied Data Science and Artificial Intelligence

Jan. 2023 - Dec. 2025

- Graduated *Cum Laude*
- Thesis: *Towards Flexible and Expressive Generative Models for Tabular and Relational Data* 📄
- Advisor: Luca Bortolussi (University of Trieste), co-advisor: Sebastiano Sacconi (Aindo SpA)
- Developed new techniques for tabular data generation based on diffusion models, graph neural networks, neuro-symbolic constraints and probabilistic circuits. PhD in collaboration with Aindo SpA.
- Teaching assistant for the Master's course in Probabilistic Machine Learning (2024).
- Reviewer for QEST 2023, AAAI 2026 and NeurIPS 2026 conferences.

School of Informatics, University of Edinburgh

Edinburgh, UK

Visiting Researcher

May. 2025 - Oct. 2025

- Research on tabular data generation with probabilistic circuits. Supervisor: Antonio Vergari.

University of Trieste, International School for Advanced Studies (SISSA)

Trieste, Italy

Master of Science (MSc) in Data Science and Scientific Computing

Oct. 2018 - Dec. 2020

- Graduation score: 110/110 *Cum Laude*
- Thesis: *An Information-Theoretic Investigation Into Epigenetic Regulation of Gene Expression* 📄

University of Brescia

Brescia, Italy

Bachelor of Engineering (BEng) in Computer Science and Engineering

Sept. 2015 - Sept. 2018

- Graduation score: 110/110 *Cum Laude*

Work Experience

Aindo SpA

Trieste, Italy

Research Scientist

Jan. 2023 - May 2026

- Developed a novel privacy evaluation method for synthetic data and benchmarked existing privacy metrics.
- Conducted research on generative models for tabular and relational data (PhD collaboration).

Trality GmbH

Vienna, Austria

Applied Scientist, Backend Developer (quantitative finance)

May. 2021 - Dec. 2022

- Back-end development: Implemented trading strategy optimizer, statistical indicators, rebalancing of margin-trading portfolios.
- Research: Conducted research on cryptocurrency time series prediction models, reinforcement learning for cryptocurrency trading, performance metrics for trading strategies.

Deloitte Consulting

Milan, Italy

IT Analyst

Feb. 2021 - Apr. 2021

- Performed statistical analysis of banking datasets and defined automated data quality controls.
- Analyzed data governance solutions in the banking sector.

Aindo SpA

Trieste, Italy

Research Intern

Sept. 2019 - Dec. 2019

- Developed a deep generative model for multivariate financial time series simulating real statistical properties.

Skills and Interests

Programming	Python (Advanced), C++, C, SQL, R, Java (Familiar)
Tools & Frameworks	PyTorch, Pyro, Git, LaTeX, shell scripting, Docker, python scientific libraries (NumPy, SciPy, Pandas, scikit-learn)
Areas of Interest	Probabilistic models, machine learning & AI, software engineering, generative AI
Languages	English (fluent, certified C1), Italian (native)

Honors & Awards

2024	Best Poster , Eastern European Machine Learning Summer School (EEML)	Novi Sad, Serbia
2018	Data Science Excellence Fellowship (12k€) , International School of Advanced Studies (SISSA)	Trieste, Italy

Publications

	ReMIA: a Powerful and Efficient Alternative to Membership Inference Attacks Against Synthetic Data Generators	arXiv
	Scassola D., Coser A., Saccani S.	2026
	A Sobering Look at Tabular Data Generation via Probabilistic Circuits	UAI
	Scassola D., Ponsford D. (equal contribution), Javaloy A., Saccani S., Bortolussi L., Gouk H., Vergari A.	2026
	Graph-Conditional Flow Matching for Relational Data Generation	AAAI
	Scassola D., Saccani S., Bortolussi L.	2026
	Zero-Shot Conditioning of Score-Based Diffusion Models by Neuro-Symbolic Constraints	AAAI
	Scassola D., Carbone G., Saccani S., Bortolussi L.	2025
	Multi-Objective Reward Generalization: Improving Performance of Deep Reinforcement Learning for Selected Applications in Stock and Cryptocurrency Trading	Neural Computing & Applications
	Cornalba F., Disselkamp C., Scassola D. (equal contribution), Helf C.	2024
	Model Abstraction and Conditional Sampling with Score-Based Diffusion Models	QEST
	Bortolussi L., Cairoli F., Giacomarra F., Scassola D.	2023